BLAISE XTECH PHASE I / IDEA COMPETITION

SpectraPass

Point-of-Use Material Authentication for Industrial MRO

SELECT SPECTRAPASS to turn Blaise into an uncertainty-aware material passport that helps technicians verify the right industrial material before use - not after a costly mix-up.

1. PROBLEM

Industrial maintenance and manufacturing teams handle visually similar lubricants, adhesives, sealants, coatings, polymers, and other consumables. A mislabeled, substituted, or counterfeit material can create rework, downtime, traceability gaps, and lab escalation. Labels and purchase records establish provenance, but do not independently verify the material in a technician's hand.

2. SOLUTION

Pair a Blaise Raman scan with the facility's approved-material library. SpectraPass preprocesses a spectrum, compares it with versioned reference fingerprints, then returns MATCH, MISMATCH, or INCONCLUSIVE with calibrated confidence, nearest-reference evidence, and QR/lot/device metadata. An offline-first workflow preserves point-of-use checks. Uncertain or out-of-distribution scans abstain and route to existing verification procedures.

3. UNIQUENESS

Provenance-oriented AI, not generic chemical identification. Each organization keeps a narrow library of materials it actually authorizes. Similarity retrieval provides inspectable nearest-reference evidence; uncertainty calibration and explicit abstention reduce false certainty. Versioned reference sets, drift checks, and lot/device metadata make each decision reproducible and auditable.

4. IMPACT + SCALE

Start with industrial MRO and manufacturing teams that repeatedly verify incoming or point-of-use materials. The inference core can scale across facilities while each organization controls its approved references and SOPs. The goal is to catch identity mismatches before use, reduce avoidable lab escalations, and preserve existing quality controls rather than replace them.

5. FEASIBILITY

Phase II starts with a bounded reference set and repeatable scan SOP. Collect replicate spectra across known lots and operating conditions; establish classical similarity / PLS / SVM baselines; compare a compact 1-D neural encoder only if it improves held-out performance. Add reference-library sync, QR/lot capture, offline inference, calibration, and audit export. Validate with lot-held-out splits plus explicit unknown-material tests.

6. TEAM READINESS

Team: TokenJunkieLabs / Commons. Engineering-first execution: versioned data pipelines, deterministic evaluation, compact AI inference, reproducible artifacts, and rapid software prototyping. Exact entrant/contact details remain OWNER_PASTE_REQUIRED. If selected, add a Raman/materials domain advisor before interpreting spectral performance beyond the validated task.

7. SUCCESS METRICS

Measure held-out known-material identification, false-accept rate on wrong/unknown materials, calibration error, abstention rate, scan-to-result latency, repeatability across replicate scans/lots, technician task-completion time, verified scans per active site, and audit-record completion. Numeric thresholds are frozen only after the first Blaise dataset establishes realistic baselines.

8. COMMERCIALIZATION

Plan dual-store distribution as Blaise SDK support permits, with B2B subscriptions for site libraries, audit retention, team administration, and enterprise integrations. Bottom-up operating target: **250 organizations x 4 active sites x $85/site/month = $1.02M ARR**, before enterprise API/custom-library upsells. This is a go-to-market target, not current revenue or a market-size claim.

TEAM TokenJunkieLabs / Commons	**PRIMARY CONTACT / EMAIL / AFFILIATION / CITY / COUNTRY** OWNER_PASTE_REQUIRED
VIDEO Required, <= 3:00. Storyboard ready in Commons.	**STATUS** Draft only - not registered, not submitted, no award/payment claimed.

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